OPEN SOURCE · AGPL-3.0

Competitive Intelligence Dashboard

Blockstream operates across seven product verticals and competes with over 30 organisations. Today, tracking competitors means checking GitHub, news, and app stores manually — once a week, if at all.

The Problem

- → Competitor activity is scattered across GitHub, news, and app stores
- → Manual monitoring doesn't scale to 30+ entities across 7 verticals
- → Signals are noticed days late — or not at all
- → No shared view of what changed and why it matters

The Goal

- → Aggregate public signals automatically, on a schedule
- → Alert when a metric moves beyond a meaningful threshold
- → Surface new competitor candidates from news
- → Deliver a weekly intelligence briefing via AI

Built by **Carlo Cogni**

SYSTEM DESIGN

Three-Layer Architecture

One codebase · one process · one database. No microservices, no message broker.

LAYER 1

Data Ingestion

- → GitHub API
- → Google News RSS
- → App Store API (iOS)
- → Google Play (scraper)
- → Twitter v2 API
- → Reddit public API
- → GitHub releases.atom

7 fetchers · each independently testable · failures never crash the scheduler

→

LAYER 2

Storage & Detection

- → SQLite (zero-setup, production schema)
- → MetricSnapshot (raw JSON) + flat Metric rows
- → Delta detection — static & Isolation Forest
- → Compound rules via Genetic Algorithm
- → Competitor candidate extraction (LLM)
- → APScheduler — per-source configurable intervals

→

LAYER 3

Dashboard

- → Entity overview (sortable table)
- → Vertical drill-down with breadcrumbs
- → Head-to-head comparison + radar charts
- → Unified feed (News / Releases / Alerts)
- → Alert feed with HTMX live polling
- → AI Observatory & weekly briefings

Django 6.x · SQLite · APScheduler · HTMX 1.9 · Alpine.js 3 · Tailwind CSS · Chart.js 4 · scikit-learn · Ollama / Groq · uv · pytest + factory_boy

⚡ NOT IN THE BRIEF — ADDED ANYWAY

AI Intelligence Layer

Six interconnected AI systems sit on top of the three-layer pipeline, transforming raw signals into actionable intelligence — none of this was required by the assignment.

Isolation Forest anomaly detection · Genetic Algorithm rule optimizer · LLM competitor extraction · News sentiment & signal classification · Ollama + Groq weekly briefings · Momentum speed & acceleration tracking

WHAT IT DOES

Demo Results

- 7 verticals live
- 31 tracked entities
- 7 data sources
- Login: batman / the hero we deserve

Entity Overview
Sortable table of all 31 entities across 7 verticals. Filter by vertical, search by name, toggle between watchlist and full scope.

Entity Drill-Down
Company → Product navigation with breadcrumbs. Sparkline charts (30d / 90d / All) for every metric with delta badges.

Head-to-Head Comparison
Select any two entities. Radar chart + side-by-side metrics + trend lines. Save comparisons to revisit later.

Unified Feed
Three-column layout: News articles · GitHub releases · Alerts. Filter by entity. Auto-refreshes via HTMX polling.

Alert System
Fires when a metric moves beyond a configurable threshold. Supports METRIC_SPIKE, NEW_RELEASE, NEW_REPO, COMPETITOR_NEW, and COMPOUND_SPIKE types.

Competitive Landscape Map
Force-directed graph of all 31 entities. Node size = composite score. Three edge types: news co-mentions, shared GitHub org, correlated metric trajectories. Click any node to drill down.

AI Intelligence Layer — beyond the assignment brief

Six AI systems that were not requested but were built anyway. They transform raw signals into actionable, self-improving intelligence.

Smart Anomaly Detection
Learns each entity's normal metric behaviour and alerts only on statistically unusual changes — not arbitrary fixed thresholds.

Isolation Forest · scikit-learn

Compound Alert Rules
Discovers multi-metric patterns that correlate with meaningful competitive events by evolving rules from user dismissal feedback.

Genetic Algorithm · F-beta fitness

Competitor Detection
Scans every news article for untracked entity mentions. LLM determines if a name is a genuine competitor. Fires alert after 3 mentions.

LLM structured output · Ollama / Groq

News Intelligence
Classifies every article by sentiment and strategic signal type. Generates summaries and key quotes. Plotly heatmap reveals media coverage patterns across entities and weeks.

LLM enrichment · Ollama-only · Plotly.js

Weekly Intelligence Briefings
Generates a narrative summary of the week's most important competitive signals — on-demand or on a schedule, with email delivery.

Ollama (local) · Groq cloud fallback

Momentum Tracking
Aggregates all metric velocities into a single speed score per entity. Tracks acceleration to spot rising stars and fading leaders.

Z-score normalization · quadrant analysis

IMPLEMENTATION STATUS

Built vs. Documented for Later

FEATURE	STATUS	NOTES
7 data fetchers (GitHub, News, iOS, Android, Twitter, Reddit, Changelog)	BUILT	All independently testable, mocked test suite, failures isolated
Delta detection — static thresholds + Isolation Forest anomaly detection	BUILT	Per-entity, per-metric models; cold-start LLM suggestions
⚡ AI LAYER — 5 SYSTEMS, NOT REQUESTED, BUILT ANYWAY		
Anomaly detection — Isolation Forest per (entity, metric)	AI BONUS	Context-aware thresholds; cold-start LLM suggestions for new entities
Compound alert rules via Genetic Algorithm optimizer	AI BONUS	Evolves multi-metric rules from user dismissal feedback (AlertSeen labels)
Competitor candidate detection — LLM entity extraction from news	AI BONUS	Structured LLM output; promote/dismiss workflow; COMPETITOR_NEW alerts
News Intelligence — LLM sentiment & signal classification	AI BONUS	Per-article sentiment, signal type, summary, key quote; Plotly heatmap; on-demand takeaways; Ollama-only mode
Weekly intelligence briefings (Ollama local + Groq cloud fallback)	AI BONUS	On-demand or scheduled; email delivery; provider toggle in UI
Momentum tracking — speed & acceleration per entity	AI BONUS	Z-score composite; scatter quadrant map; filterable time-series; entity badge
Full dashboard — overview, drill-down, head-to-head, feed, alerts, candidates	BUILT	HTMX live updates, Chart.js sparklines, Alpine.js interactions
Competitive Landscape Map — interactive force-directed entity graph	BUILT	Cytoscape.js; 3 edge types: co-mention (news), shared GitHub org, correlated metric trajectories ($r \geq 0.7$)
CoinGecko / Mempool.space market metrics	DOCUMENTED	Free APIs, no keys; add as two new fetchers
App Store & Play Store version string display	BUILT	iOS and Android version badges on entity detail page, read from MetricSnapshot
Job changes / hiring tracking	DOCUMENTED	No free API; use optional careers_rss_url in entities.json

ROADMAP

Next Steps

Three highest-ROI improvements, ordered by impact vs. effort.

Market Metrics (CoinGecko + Mempool) 1

Add Bitcoin price, market cap, and network hash rate alongside competitive metrics. Both APIs are free with no keys required. Wires directly into the existing fetcher and delta pipeline — no architectural changes needed.

↑ Strategically relevant for Blockstream · Low effort (~2 days)

Cross-Source Alert Corroboration 2

Group alerts that fire within 48 hours across multiple sources for the same entity into a single "corroborated signal." Reduces noise, increases confidence. No new data needed — purely a post-processing layer on existing alerts.

↑ Highest noise-reduction value · Medium effort (~3 days)

AI-Enhanced Momentum 3

The momentum system currently uses pure statistics (z-scores). Three AI enhancements are ready to build: **anomaly-aware weighting** (Isolation Forest scores amplify unusual signals), **GA-learned metric weights** (evolve optimal weights from user engagement), and **LLM trajectory narratives** ("Green Wallet is accelerating due to a coordinated product push"). Each plugs into existing infrastructure.

↑ Transforms numbers into strategic intelligence · Medium effort (~3-5 days)

Repository
github.com/CarloCogni/Competitive-Intelligence-Platform

Quick start
Unzip → double-click start.bat · http://localhost:8000

Login
batman / the hero we deserve

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